

VitalQuant: A Multimodal Wearable Sensing Platform with Quantum-Inspired Readout Modelling for Early Physiological Deterioration

VitalQuant Research

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<https://github.com/Riyu-D1/VitalQuant>

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Abstract

Abstract. Sepsis and post-operative physiological deterioration remain among the deadliest time-critical conditions in medicine: each hour of delayed treatment measurably increases mortality, yet standard monitoring detects deterioration only after symptoms are florid. We present **VitalQuant**, a research platform for multimodal wearable physiological sensing built around a single design rule — *measurement honesty before clinical claims*. The system integrates (i) a custom multi-sensor wearable hardware design and ESP32 firmware with provenance-first timestamping; (ii) an idempotent store-and-forward ingest API with device authentication and a statistical clock-correction protocol; (iii) a partitioned relational data architecture separating raw, cleaned, quality-assessed, feature, and model layers, with a `data_class` provenance tag on every row; (iv) a versioned signal-processing stack (`fusion_v2`) producing pulse-rate-variability, respiratory, morphological, spectral, contact, and motion features under explicit signal-quality gating; (v) a label-free anomaly-detection layer with per-subject baselines and an evaluation protocol that declares its own synthetic-label caveat in the model record; and (vi) a quantum-inspired readout simulator with a composable noise hierarchy (shot noise, squeezing, detection loss, additive floor, proportional tissue noise, ambient leakage). Executing the platform’s experiment matrix (136 cells, 200 Monte-Carlo trials each), we measure the squeezed-readout signal-to-noise advantage falling from $2.7\times$ under ideal conditions to $\approx 1.14\times$ once tissue noise dominates — a result the system reports rather than hides. On the platform’s synthetic-device harness the PPG chain recovers heart rate with 0.12 bpm mean absolute error, and every injected-corruption window is rejected through a named physical mechanism (rail or flat fraction = 1). VitalQuant is released source-available as a scientific artefact: a research instrument, not a medical device.

1 Introduction

Sepsis — a dysregulated host response to infection — is dominated by time to treatment. Roughly 310 million surgical procedures are performed annually worldwide, and sepsis remains a leading cause of in-hospital and post-discharge deterioration; the critical-care literature estimates that each hour of delay in effective antimicrobial therapy raises mortality on the order of 7–9%, and survival drops by roughly 70% when recognition is late [1, 2]. Conventional ward monitoring — intermittent spot checks aggregated into early-warning scores such as NEWS2 — leaves a detection gap on the order of a half-day between onset of the physiological cascade (micro-circulatory failure, vasodilation, lactate rise, cytokine surge) and clinical recognition.

The premise of VitalQuant is that continuous, multimodal, quality-aware wearable sensing can move the earliest detection point forward by 6–12 hours — not by diagnosing sepsis, which remains a clinical judgement requiring laboratory evidence of infection and organ dysfunction, but by surfacing statistically anomalous departures from a subject’s *own* baseline early enough to trigger escalation and workup. The modalities we deploy map onto the physiological axes the cascade perturbs: dual-wavelength PPG morphology and perfusion index track microcirculatory tone; pulse-rate variability

tracks autonomic withdrawal; the infrared thermometer and ambient-differential features track dysregulated thermogenesis; contact-force and motion channels exist primarily to *gate* the rest — a wearable’s first failure mode is not physiology but physics.

The corresponding biomarker axes of the underlying cascade — interleukin-6 and the broader cytokine surge, rising lactate, falling pH, collapsing tissue perfusion — are mostly not directly measurable at the wrist. The platform’s position is that their *correlates* are: fever response is a slope-and-differential problem before it is a level problem; vasodilation and capillary leak appear in PPG morphology (reflection index, crest time, area ratio) and perfusion index before they appear in blood pressure; autonomic withdrawal appears in PRV band structure before sustained tachycardia. This is the framing the feature layer (Section 7) is built around.

Building such a system credibly is substantially harder than building a sensor. Wrist PPG is corrupted by motion and contact artefacts; battery-device streams arrive out of order with clock drift, replay, and firmware churn; labels for “early sepsis” are scarce and contestable; and a novel optical readout — quantum-enhanced detection — carries an honest-uncertainty budget that must be modelled, not asserted. VitalQuant therefore treats the *measurement platform itself* as the research ob-

ject: provenance, quality gating, and simulated-vs-real separation are encoded in schema constraints and code rather than in policy documents [19].

Contributions.

- (1) A documented hardware–firmware–cloud architecture for a multimodal wearable (dual-wavelength PPG, 11-channel spectrophotometry, IR thermometry, IMU, contact force, environment) with a two-generation sensor roadmap and per-batch hardware-profile hashing.
- (2) An ingest protocol in which exactly-once batch delivery, clock-anchor provenance, and a `data_class` tag (real/synthetic/simulated) are enforced at the schema and validator level rather than by convention.
- (3) The `fusion_v2` processing stack: SQI-gated PPG analytics (eight-component multiplicative quality index, matched-filter template rescue, cross-channel z-score fusion), PRV, respiration, morphology, and contact/motion confidence — exercised end-to-end on the platform’s synthetic device.
- (4) A label-free anomaly layer (Isolation Forest + PCA reconstruction + lag forecaster, median/IQR-normalised, max-aggregated) with per-subject median/MAD baselines, leave-one-session-out evaluation, and a mandatory honesty field in the model spec.
- (5) A quantum-inspired readout simulator whose executed experiment matrix quantifies the squeezed-light advantage boundary under tissue and ambient noise ($2.7\times \rightarrow 1.14\times$), with simulated outputs structurally separated from real data.

Scope. VitalQuant is a research instrument. Nothing herein is a diagnostic claim; model outputs are experimental anomaly scores. Section 14 states limitations and self-imposed evaluation boundaries.

2 Related Work

Wearables for deterioration. Wearable-derived heart-rate and variability features have been linked to infection risk and clinical deterioration, most visibly in pre-symptomatic COVID-19 detection from smartwatch data [3, 4]. Most work post-processes exports from closed devices; few publish the acquisition-to-feature stack with artefact accounting. Our contribution is infrastructural: quality, contact, and clock state are recorded beside every measurement so downstream consumers weight rather than blindly average.

PPG quality and pulse analysis. SQIs fusing spectral, morphological, and template evidence are established practice [5, 6, 7]. Our index is multiplicative with hard gates — rail saturation or flatlining caps the composite — because in ambulatory data a single catastrophic window is the norm, not the tail.

Label-free anomaly detection. Isolation Forests, reconstruction-error models, and forecast-residual detectors are the standard response to scarce labels [8]. We pair them with per-subject baselines (“self as control”), which is well motivated in ambulatory physiology where inter-subject dispersion dwarfs within-subject change.

Dataset	Validates
BUT PPG	SQI vs. expert quality labels
PPG-DaLiA	wrist PPG HR vs. ECG truth
WESAD	multimodal baseline deviation
Capnabase / BIDMC	resp. + SpO ₂ transfer
PulseDB	BP-adjacent PPG morphology
MIMIC-III/IV	clinical deterioration cohorts
WildPPG	field-motion robustness

Table 1: External dataset registry and the subsystem each one validates.

Quantum-enhanced optical sensing. Squeezed states reduce photodetection shot noise below the classical limit; gravitational-wave observatories routinely gain several dB of effective sensitivity this way [9, 10], and the technique has been proposed for biological photometry and Raman spectroscopy where photon budgets are damage-limited [11, 12]. Whether the advantage survives biological noise is the question our simulator quantifies.

Provenance-aware time-series infrastructure. Clinical waveform stores (MIMIC-derived pipelines, PhysioNet tooling) generally treat provenance as metadata convention rather than a type-level guarantee; the open-hardware wearable literature similarly describes firmware, ingest, and analysis in separate artefacts with hand-shaken contracts. VitalQuant’s ingest envelope, clock-anchor protocol, and `data_class` column are an attempt to make the provenance constraint machine-checkable: a simulated value cannot be stored as real, an unknown channel cannot enter the schema, and a replayed batch cannot double-count.

Datasets. Validation builds on BUT PPG [13], PPG-DaLiA [14], WESAD [15], MIMIC-III/IV and PulseDB [16], Capnabase/BIDMC [18], and WildPPG [17]; Table 1 maps each corpus to the subsystem it validates. A dedicated `vitalq-import` importer maps external channel conventions onto the registry (e.g. `WFDB PLETH→ppg.ir`, `RESP→resp.waveform`, `II/V/AVR→ecg.*`), mints deterministic uuid5 session and batch IDs so re-imports are idempotent, and honours the ingest rate limiter with backoff on 429 — so external corpora enter the system through the same contract as the hardware, including the same `data_class` tagging.

3 System Architecture

The platform is layered so that each higher layer can assume — and explicitly check — guarantees made below (Figure 1). The monorepo contains firmware and electrical design notes, the ingest service, the relational schema, the processing worker, the ML layer, the quantum simulator, the synthetic device generator, a Streamlit research dashboard, and the project website.

L6 Dashboard/Web — research UI, SIMULATED banners
L5 ML — baselines, anomaly ensemble, experiments
L4 Processing — SQL-gated features (<i>fusion_v2</i>)
L3 Storage — <i>raw/clean/quality/</i> <i>features/ml/config/meta</i> schemas
L2 Ingest API — batches, clock sync, auth
L1 Firmware — drivers, ring buffer, uploader
L0 Hardware — multisensor wearable + power

Figure 1: Layered architecture. Guarantees are checked upward, not assumed.

Function	hw_v0	hw_v1	Rate, bus
PPG	MAX30102	MAX86141	200 Hz, SPI
Spectral	AS7341	AS7341-DLGM	2 Hz, 0x39
Temp. (object)	MLX90614	MLX90637	1 Hz, 0x3B
Environment	BME280	BME280	0.5 Hz, 0x76
Motion	MPU6050	ICM-42670-P	52 Hz, 0x68
Contact	FSR402	FSR402	20 Hz, ADC1
Buffer	SD card	flash ring 1800 s	—

Table 2: Sensor matrix (I²C addresses for hw_v1). Power tree: MCP73831 charge controller, XC6206P332MR and TPS7A2 rails, CP2102 USB–UART.

3.1 Data flow, end to end

A measurement’s life begins as an `esp_timer` reading inside a driver task, travels as a windowed sample through the flash ring into a `BatchIngest` envelope, lands in `raw` under its (`device_id`, `batch_id`) anchor with its clock anchors recorded, is resampled and gap-masked into `clean`, scored for quality and contact state into `quality`, reduced to named features in `features.windows`, referenced by `features.baselines`, and finally scored by model versions recorded in `ml.models` — at every hop carrying `data_class`, `pipeline_version`, and window-level source IDs. Nothing in the chain is allowed to forget where it came from; that is the entire honesty argument in one sentence.

3.2 Sensing hardware

Two generations are specified as versioned YAML profiles; a canonical-JSON SHA-256 of each profile (`profile_hash`) travels with every batch, binding measurements to the exact configuration that produced them. `hw_v0` is a breakout rig for firmware bring-up; `hw_v1` is the custom PCB (Table 2).

The `hw_v1` PPG upgrade is substantive: MAX86141 delivers simultaneous red (660 nm) and IR (880 nm) channels with ambient-light cancellation (ALC) at 200 Hz on a dedicated SPI bus, freeing the shared I²C bus and enabling the dual-wavelength ratio underlying the uncalibrated SpO₂ proxy. Simultaneous red/IR acquisition matters more than it appears: the *R* ratio assumes both wavelengths sample the same cardiac phase; interleaved channels inject a half-cycle timing error directly into $(AC/DC)_{\text{red}}/(AC/DC)_{\text{ir}}$. The 200 Hz rate is chosen for morphology — rise time and crest time features need $\gtrsim 100$ Hz to localise the systolic foot to within a few millisec-

onds — and is compatible with the sub-5 ms cross-channel clock target.

The spectral front-end reads 11 channels (f1–f8 spanning the visible band, clear, NIR, flicker) in two parallel read groups at gain 64 with 27.8 ms integration, alternating emitter-on and dark frames so ambient leakage is subtracted rather than assumed — the same accounting discipline as the quantum simulator’s ambient term, applied to classical hardware. The two read groups introduce μs -level intra-frame offsets that the wire format records explicitly (`read_group` per frame), keeping frame timestamps honest rather than treating an 11-channel vector as simultaneous. The IMU moves to ICM-42670-P (MPU6050 is not recommended for new designs) at 52 Hz — sufficient for the 0.5–4 Hz PPG-band motion estimate with headroom — and the FSR supplies an *ordinal* contact level used only for on-skin gating, deliberately not a calibrated pressure reading. The object-temperature channel (MLX90637, 1 Hz) is medical-grade IR thermometry; BME280 logs ambient temperature/humidity/pressure at 0.5 Hz so object-minus-ambient features can separate true skin-temperature change from room drift.

The power tree (MCP73831 charge controller, XC6206P332MR and TPS7A2 rails, CP2102 USB–UART bridge) is sized for the combined LED drive bursts of the PPG and spectral emitters; the 30-minute flash ring on `hw_v1` absorbs connectivity gaps without losing samples. Longer-term optics work evaluates thin-film lithium-niobate modulation as a path to squeezed-light readout with a claimed 2–5 $\times$ SNR improvement in shot-noise-limited regimes; Section 9 reports the simulated boundary of that claim.

4 Firmware and Clock Discipline

Firmware drivers implement a uniform contract — `init(cfg)` applies the hardware profile, `read(out,max)` returns timestamped samples, `healthy()` reports liveness — and timestamps are taken *in the driver task* on the device monotonic timer (`esp_timer`, μs), never reconstructed on transmit. This rule eliminates transmit-time timestamping, which smears sample times by queue depth and corrupts interbeat statistics: a 30 ms queuing delay is 5% of a 600 ms IBI, enough to dominate the RMSSD the feature layer computes. Batching is likewise disciplined: windows carry (`channel`, `rate_hz`, `t_start`, `samples[]`) rather than per-sample timestamps, so a 200 Hz channel emits a bounded number of segments per batch while retaining μs provenance.

A `ClockService` anchors the monotonic base to SNTP wall time; a flash `RingBuffer` provides store-and-forward (1800 s on `hw_v1`, sized so a full-rate multi-channel write fits the buffer with margin); a `BatchUploader` replays batches idempotently and applies server-returned clock corrections prospectively. Because the ring is the authoritative buffer, uplink loss degrades freshness, never integrity.

Each batch carries clock anchors (monotonic μs paired with wall time). Server-side, a piecewise-linear `ClockModel`

maps monotonic to UTC between anchors,

$$t_{\text{utc}} = w_i + (m - m_i)(1 + d), \quad d \text{ in ppm}, \quad (1)$$

then fits a global drift d per device and flags jumps > 500 ms as `clock.jump` device events rather than silently re-aligning — a jump is a fact about the device (reboot, NTP step, battery swap), not something to hide. The ingest response returns `clock_correction{offset_us, drift_ppm}` so the device applies the correction prospectively rather than rewriting history. Design target: sub-5 ms cross-channel agreement, the scale at which dual-wavelength ratios and motion-PPG correlation remain meaningful.

5 Ingest API and Batch Protocol

The FastAPI service exposes session lifecycle (POST `/v1/sessions`, `/v1/sessions/{id}/end`), `/v1/subjects`, `/v1/sessions/{id}/labels`, `/v1/ingest/batch` (202), `/v1/events`, read paths for `signals/scalars/quality/features/predictions`, plus `/v1/health` and `/v1/version`. The batch envelope:

- `scalars[]` — (channel, $t_{\mu s}$, value) samples;
- `windows[]` — (channel, rate_hz, t_{start} , `samples[]`) with validators $0.01 \leq \text{rate} \leq 10^4$ Hz and $|\text{samples}| \leq 2 \times 10^4$;
- `frames[]` — spectral frames: ($t_{\mu s}$, read_group, gain, integ_ms, illumination, channels[11], T_{sensor});
- `events[]` — device events (faults, clock jumps, reboots);
- `clock` — anchors for the batch;
- `batch_id`, `session_id`, `firmware_version`, `profile_hash`.

Exactly-once replay. The row (`device_id`, `batch_id`) is the transaction anchor: the batch header is inserted first inside the payload transaction, and a duplicate key yields 409 `batch.replay` with the original outcome embedded. The semantics are worth stating precisely: a replayed batch returns the first commit’s result (acceptance, clock flags, correction) without rewriting payload rows, so at-most-once transport plus idempotent server equals exactly-once effective delivery. The synthetic device emits deterministic uuid5 batch IDs exercising precisely this path — the same content resubmitted produces the same ID, so even generator restarts replay safely.

Fail-closed validation. Channels are resolved against a registry (`ppg.red`, `ppg.ir`, `ppg.ch3/ch4`, `temp.object`, `temp.ambient`, `env.*` (temperature, humidity, pressure, gas_resistance), `motion.accel_x/y/z`, `motion.gyro_x/y/z`, `contact.level`, `sys.battery_v`, `resp.waveform`, `ecg.ii/v/avr`) — unknown channels are rejected with `schema.channel`, not coerced. The payload cap is 2 MB (`payload.too_large`), batches must be non-empty, window rates are bounded to $[0.01, 10^4]$ Hz, and every list field is length-capped so a malformed batch cannot allocate unbounded work. Validation happens before any write; a rejected batch leaves no residue.

Auth and fairness. Devices authenticate with

Code	Meaning
<code>auth.invalid</code>	bad/absent device key
<code>auth.revoked</code>	key disabled
<code>batch.replay</code>	idempotent duplicate
<code>schema.*</code>	envelope/field violations
<code>clock.*</code>	anchor/model failures
<code>payload.too_large</code>	>2 MB body
<code>rate.exceeded</code>	token bucket
<code>session.state</code>	wrong lifecycle phase

Table 3: Error taxonomy: stable codes let firmware react programmatically.

X-Device-Key (`vq_` + 24-byte URL-safe token, shown once at issuance by the `vitalq-device` CLI; only its SHA-256 is stored in `meta.devices.api_key_hash`). Revocation yields 403; a wrong key yields 401 with no oracle about why. A token bucket (default `VITALQ_RATE_RPS=50` req/s per device) rate-limits write paths (`rate.exceeded`); responses carry stable machine codes (`auth.*`, `schema.*`, `clock.*`, `session.state`, `batch.replay`) so firmware can branch programmatically. Human auth mode supabase exists but returns 501 until JWKS verification is wired — failing closed by default rather than shipping a half-verified identity layer.

6 Data Architecture

PostgreSQL 17 (Supabase migrations) is organised into seven schemas: `config` (hardware profiles, devices), `raw` (append-only ingest), `clean` (resampled streams), `quality` (SQL, contact state), `features` (windows, baselines), `ml` (experiments, models, predictions), and `meta` (subjects, labels).

Provenance as constraint. Every fact table carries `data_class ∈ {real, synthetic, simulated}` NOT NULL; synthetic and simulated output cannot contaminate real rows — the contamination error class is removed by construction.

Storage shaped to the workload. Waveform segments store samples as a native `real[]` array per window ($\sim 100\times$ write reduction vs. row-per-sample, and contiguous arrays compress far better than heap rows); raw tables are PARTITION BY RANGE monthly with an `ensure_monthly_partitions()` helper so ingest never blocks on partition creation (TimescaleDB hypertables were evaluated and dropped for Supabase PostgreSQL 17 compatibility); raw is append-only, so corrections and re-ingests arrive as new facts and history is never rewritten. `meta.subjects` is pseudonymised with an opaque subject code plus `consent_ref` pointing at the consent artefact; `meta.labels` records label provenance (who or what produced the label, and when); `features.windows` stores values as JSONB with `feature_set='fusion_v2'` and `source_window_ids`, so every feature is traceable to the exact windows that produced it and any reprocessing is auditable; `features.baselines`

holds per-subject median/MAD over a 14-day rolling window; `ml.experiments`, `ml.models`, and `ml.predictions` link predictions to the model version and experiment that produced them.

Truth vs. judgement. `quality.channel_quality` stores SQI and its component vector (`sqi_detail` JSONB) plus `pipeline_version`, so pipeline versions can be compared rather than conflated; `quality.contact_state` stores (`contact_quality`, `motion_score`, `sensor_confidence`, `on_skin`) per window; `clean.signal_windows` carries `resample_hz=100` and a 0/1/2 `gap_mask` distinguishing measured samples, short interpolations, and genuine gaps. `ml.predictions` stores `kind='anomaly_score'` — the schema has no diagnosis concept. RLS policies are planned so subject-facing reads never join raw device traffic.

7 Signal Processing

The worker executes eight stages: **S1** ingest read → **S2** contiguous-run streaming (50 ms gap tolerance) → **S3** fixed feature-grid windowing → **S4** clean-layer resample to 100 Hz with a 0/1/2 gap mask → **S5** per-channel chains → **S6** cross-channel fusion → **S7** session-level aggregates on the concatenated fused stream → **S8** idempotent writes (ON CONFLICT DO NOTHING) stamped with `pipeline_version` and `data_class`.

7.1 PPG chain (v2)

Per window $x[n]$ at f_s : (1) *condition* with a 0.5–4 Hz fourth-order Butterworth bandpass (`sosfiltfilt`, zero-phase — the band covers 30–240 bpm with harmonics, rejecting baseline wander and high-frequency noise); (2) *detect beats* by prominence gating at 0.5σ of the filtered signal with a 0.4 s refractory and 2.0 s max IBI, then *refine onsets* to foot minima in a neighbourhood around each peak so IBIs are anchored to the physiological onset rather than the amplitude peak; (3) *clean IBIs* by rejecting intervals with $|ibi - ibi| > 0.20$ s ($\sim$ = window median — a fixed MAD-style gate that resists exactly the bursts of ectopic-looking artefacts PPG produces); (4) compute *PRV* — `hr_bpm`, `RMSSD`, `SDNN`, `IBI CV` — deliberately labelled pulse-rate variability, since wrist PPG times pulse arrival, not cardiac electrical activity; a 4 Hz-resampled tachogram yields Welch band powers VLF (0.003–0.04 Hz), LF (0.04–0.15), HF (0.15–0.4) in ms^2 , the LF/HF ratio, and an independent PRV-domain respiratory estimate `resp_hz_prv`; (5) *morphology*: pulse amplitude, rise time, 50% width, crest time, area ratio (systolic vs. diastolic integral split), and reflection index — the vascular-tone axes sepsis perturbs first; (6) *respiration*: Hilbert-envelope RIAV within a fixed 0.10–0.60 Hz band on ≥ 20 s windows, emitted with its own `resp_confidence` rather than merged blindly with the PRV estimate.

Quality index. The SQI is assembled from eight per-window components: distribution shape (skewness, kurtosis), artefact detectors (flat fraction — share of samples at zero

derivative; rail fraction — share clipped at the ADC rail), spectral purity (dominant-band share of total power), periodicity (normalised autocorrelation first-sidelobe height), template correlation against a self-derived beat prototype, and perfusion index $PI = AC/DC$. Components combine *multiplicatively*, and hard gates cap the composite: flatlining multiplies by 0.3, railing by 0.4. The multiplicative form is the point: additive scores let strong components compensate for a catastrophic one; here, a railed or flatlined window cannot smuggle a passing average no matter how periodic the remainder looks.

Template rescue and fusion. Where prominence gating under-counts (low-perfusion or noisy-but-periodic windows), a matched-filter detector (normalised cross-correlation ≥ 0.5 against the window’s own beat template) emits `tm_hr_bpm` and `tm_n_beats`; when the template path finds $> 20\%$ more beats, the window’s `hr_source` is marked `template`, keeping the provenance of the HR estimate inspectable. Co-windowed red/IR channels are fused as the SQI-weighted z-score combination $\hat{z} = \sum_i SQI_i z_i / \sum_i SQI_i$, so a clean channel carries a noisy one rather than vice versa, and the ratio $R = (AC/DC)_{\text{red}} / (AC/DC)_{\text{ir}}$ yields a strictly labelled *uncalibrated* SpO₂ proxy 110–25R — the classical linear model, deliberately uncalibrated: we publish it internally as research telemetry, never as a reading.

7.2 Session-level aggregates

Per-window features are concatenated across contiguous runs and re-estimated at session level (`sess_` prefix) on the fused stream, so PRV and respiration statistics see long baseline structure that a single window cannot. Flags derived at session level make chronic-vs-acute distinctions visible that windowed flags cannot.

7.3 Scalar, contact, and spectral chains

Temperature features cover level, short-window std, slope ($^{\circ}\text{C}/\text{min}$), slew, object-minus-ambient differential, and deviation from a rolling personal baseline — fever response is a slope-and-baseline problem before it is a level problem, and the ambient differential suppresses room-drift confounds. Environment features (ambient temperature, humidity, pressure) feed the differential terms rather than the classifier directly.

Contact quality is mapped from the FSR’s seated band (≈ 1500 counts seated, < 50 treated as off-skin); a two-state hysteresis (enter-off after 2 consecutive low windows, exit-on after 2 consecutive high) debounces `on_skin` so a brief strap bump doesn’t oscillate the state. Motion score = $\text{clip}(0.6 b_{\text{band}} + 4j, 0, 1)$ where b_{band} is accelerometer energy in the PPG band (0.5–4 Hz — the frequencies that masquerade as pulse) and j is jerk; spectral frames are dark-frame subtracted, CLEAR-normalised, and reduced to fixed ratios (`f5/f3`, `NIR/vis`) that are robust to overall intensity scale. The three gates multiply into

$$\text{sensor_conf} = cq(1 - \text{motion})(\text{on_skin} ? 1.0 : 0.2), \quad (2)$$

so off-skin data is discounted to 20% rather than silently trusted or silently deleted, and honest flags (`flag_tachy`

> 100 bpm, $\text{flag_brady} < 45$, $\text{flag_low_sqi} < 0.2$, flag_contact_off) make every threshold decision inspectable in the database and the dashboard.

8 Machine Learning Layer

A maturity ladder organises the layer: **L1** quality, **L2** features, **L3** label-free anomaly detection, **L4** risk — with L4 blocked until honest labels exist.

Baselines and ensemble. Per-subject median/MAD statistics over a 14-day window seed `features.baselines`, giving each subject a robust reference distribution per feature before any model is fit — MAD is used instead of the standard deviation specifically because baseline computation must not be dragged by the very anomalies it exists to detect. A personal-baseline Isolation Forest (100 trees, contamination 0.05, robust scaling) scores each window and writes predictions carrying `top_deviant` — the feature furthest from baseline — so every elevated score states its reason rather than arriving as a bare number.

The ensemble adds two detectors covering different failure modes: a PCA reconstruction-error model (95% variance retained, ≤ 10 components; catches correlation-structure breaks that axis-aligned detectors miss, e.g. HR-motion decorrelation) and a ridge forecaster over lagged windows (≤ 3 lags; catches dynamics that violate short-term predictability even when marginals look normal). Each detector score s_d is normalised by its own running median/IQR,

$$\tilde{s}_d = \frac{s_d - \text{med}_d}{\text{IQR}_d + \epsilon}, \quad s_{\text{ens}} = \max_d \tilde{s}_d, \quad (3)$$

so detectors with different score geometries compete on a calibrated scale, and the max makes the ensemble conservative — any detector can escalate, none can veto. Each normaliser’s state is stored alongside the model record for audit.

Evaluation without pretending. The experiment harness runs leave-one-session-out (LOSO) evaluation with injected perturbations — default spec `{fused_hr_bpm :  $\times 1.6$ , temp_c :  $+3.0$ , fused_sqi :  $\times 0.1$ }` applied to a 5% prevalence of windows — reporting Mann–Whitney AUROC and average precision (AUPRC, the honest metric under class imbalance). The stored `model_spec` carries the mandatory field “*honesty: labels are injected synthetic perturbations — not real ground truth,*” and every report repeats it. Leakage controls: folds are grouped by `subject_id` (GroupKFold semantics under LOSO), a ≥ 1 -day temporal gap separates train and test windows, all statistics are fit on train folds only, and `data_class` filters keep synthetic/simulated rows out of real-data evaluations. Model provenance records the git SHA of the code that trained it.

The external plan (Phases A–E) sequences: (A) public-dataset baselines — SQI against BUT-PPG labels, HR vs. ECG MAE on DaLiA, deviation detection on WE-SAD, respiration/SpO₂ transfer on Capnabase/BIDMC; (B) synthetic-vs-real feature-distribution alignment via two-sample KS tests, gating whether the generator can stand

in during early studies; (C) personal-baseline evaluation; (D) an ablation grid over modality subsets $\{\text{PPG}\} \times \{+T, +S, +M, +C, \text{all}\}$ against {logistic regression, random forest, LightGBM} under LOSO; (E) gated sub-1M-parameter deep models, permitted only after (A)–(D) bound the classical error bars.

LAT-D-VQt-1. Separately from the open pipeline, VitalQuant works with a closed-weight clinical-decision model post-trained for the platform by LatticeAG (LoRA adapters, FP16, $8 \times A100$ -80GB) on clinical/sepsis time-series, exposing a fixed question set, calibrated confidence outputs, and wearable-only inputs. Its integration point is the `features` layer: it consumes exactly the auditable feature vectors described above, never raw waveforms, so its evidence chain is identical to the open models’. Its accuracy is unmeasured and it is documented internally as a demo until Phase-A/B evaluation catches up — stated plainly because a closed-weight component is the least auditable part of the stack.

9 Quantum-Inspired Readout Simulation

The speculative limb is a photodetection simulator answering one question: *does squeezed-light readout keep its advantage once biological noise, loss, and ambient leakage are modelled?* Signal models — a Lorentzian “Raman-like” peak (centre 1650 cm^{-1} , width 8, amplitude 1000, baseline 100, position shifting $20 \cdot \text{agg_frac} \text{ cm}^{-1}$) and an absorption dip — pass through a composable noise hierarchy:

$$\sigma_{\text{shot}} = k\sqrt{N}, \quad \sigma_{\text{sq}} = \sigma_{\text{shot}} e^{-r}, \quad (4)$$

$$V_{\text{loss}} = k^2 N (\eta e^{-2r} + (1 - \eta)), \quad (5)$$

$$V_{\text{tot}} = V_{\text{loss}} + V_{\text{floor}} + (\sigma_{\text{rel}} S)^2 + V_{\text{amb}}, \quad (6)$$

with squeezing r ($r=0.8 \Rightarrow \approx 7$ dB reduction), efficiency η , additive floor, proportional tissue noise $\sigma_{\text{rel}} S$, and ambient leakage V_{amb} . The matrix sweeps $r \in \{0, 0.25, 0.5, 0.8, 1.0\}$, $\eta \in \{0.5, 0.75, 0.9, 1.0\}$, $\sigma_{\text{rel}} \in \{0, 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$, $\text{ambient} \in \{0, 10\}$, 200 trials, fixed seed; each cell reports peak SNR, detection rate, and FPR, and the runner computes an *advantage boundary* — the regime where the best squeezed cell stops dominating classical. Outputs are `data_class='simulated'` and render only under a SIMULATED banner.

Table 4 shows the executed result (best cells). The honest headline: **the advantage collapses from $2.7 \times$ to $\approx 1.14 \times$ as proportional tissue noise dominates**. The mechanism is visible in the variance budget (3): squeezing reduces only the V_{loss} term; once $(\sigma_{\text{rel}} S)^2$ exceeds shot variance, the readout term is a small share of the total and e^{-r} barely moves it. Squeezing is therefore a conditional instrument for shot-noise-limited regimes — thin sample paths, high η , dark-adapted or shielded tissue — not a free multiplier. This is the answer the platform exists to compute: it bounds the honesty of the company’s own optical roadmap.

Tissue σ_{rel}	Ambient	SNR (class./sq.)	Advantage
0	0	33.2 / 90.1	2.72×
0	10	31.7 / 69.7	2.20×
0.001	0	33.1 / 89.8	2.71×
0.01	0	31.5 / 67.0	2.13×
0.01	10	30.3 / 57.2	1.89×
0.05	0	17.1 / 19.5	1.14×
0.05	10	16.9 / 19.2	1.14×

Table 4: Executed advantage boundary, best cell per regime ($r=0.8$, $\eta=1$): measured output of the repository’s matrix, not a hand-computed claim.

Two companion scripts broaden the picture. An amide-I Raman toy models α -synuclein aggregation fraction as a Lorentzian peak shift under Poisson detection, sweeping the $\text{SNR} \geq 3$ detection threshold against aggregation fraction to ask *how early* aggregation could be resolved in principle. A sepsis-biomarker model parameterised on published clinical data simulates logistic trajectories for IL-6, lactate, pH, and temperature with classical vs. squeezed optical detectors; its modelled detection times run ahead of the ~ 18 h symptom-onset reference by several hours. Both are illustrative demonstrations of the argument — ‘earlier detection is physically plausible’ — not evidence.

We also report a known limitation of the current build: the Monte-Carlo detection-rate statistic, which thresholds a quadratic-fit peak-centre estimator, is degenerate at the toy parameterisation (it reports ≈ 0 across cells, consistent with an estimator-resolution issue under review); the analytic SNR advantage boundary is the robust output we stand behind. This is the kind of result the simulated-vs-real separation exists for: the flaw is visible in the artefact, labelled `simulated`, and blocked from contaminating real evidence.

10 Synthetic Data and Corruption Testing

The `SyntheticDevice` emits the same `BatchIngest` wire type as hardware — it is a firmware stand-in, not a toy signal. Its PPG is a Gaussian beat train: a systolic lobe at phase 0.10 (width 0.045) plus a diastolic lobe at 0.38 (width 0.07, amplitude $0.35\times$), 2400 beats per 30 s batch, respiratory amplitude modulation ($\pm 8\%$ at 0.22 Hz), 0.1 Hz baseline wander (amplitude 300), broadband noise ($\sigma=25$), and motion bursts ($\sigma=1200$) clipped at the 45000 rail. Scalar channels follow activity-masked diurnal/random-walk series; spectral frames reproduce the two read groups with μs -level offsets and emit a dark frame every fifth cycle, exactly as `hw_v1` will. Batch IDs are uuid5-deterministic in content, so replays exercise ingest idempotency by construction rather than by test harness. A `CorruptionSpec` toggles the failure modes that actually break wearables:

Measurement	Result
PPG HR, clean synth	65.00 ± 0.21 bpm (truth 65)
PPG HR MAE	0.12 bpm
SQI, clean	mean 0.71
SQI, corrupted	mean 0.40, 50% < 0.3
Low-SQI attribution	100% rail or flat = 1.0
Quantum advantage	$2.72\times \rightarrow 1.14\times$
Test suite	35 pass / 7 skip (DB-less)

Table 5: Executed system characterisation.

Corruption	Parameter
channel dropout	<code>dropout_channel_p</code>
NTP clock jump	<code>ntp_jump_p</code> , <code>ntp_jump_s</code>
ADC rail clipping	<code>clipping_p</code>
packet loss	<code>packet_loss_p</code>
stuck sensor fault	<code>sensor_fault_p</code>
duplicate batches	<code>duplicate_p</code>

11 Results and Evaluation

We report executed measurements from the repository as it stands — every number below is the output of the platform’s own code, not an external benchmark.

PPG accuracy (synthetic ground truth). On a 120 s clean synthetic record (65 bpm, 10 s windows, 12 windows), the v2 chain recovered HR with mean 65.00 bpm, s.d. 0.21, MAE **0.12 bpm**; mean SQI 0.71. Against synthetic ground truth the bandpass–detect–clean chain is effectively unbiased; the interesting residual is variance, which the 0.21 bpm s.d. bounds.

Quality gating under corruption. With `clipping_p=0.5` and `sensor_fault_p=0.3` injected, mean SQI fell to 0.40 and 50% of windows scored below the 0.3 review threshold; *every* low-SQI window was attributable to a detected physical mechanism (rail fraction or flat fraction = 1.0). That attribution property — a rejection always carries its mechanism — is what makes the quality layer auditable rather than merely pessimistic.

Quantum boundary. Advantage $2.7\times \rightarrow 1.14\times$ as tissue noise dominates (Table 4); squeezing helps only where shot noise binds.

Suite and surfaces. The unit/integration suite is 35 passed / 7 skipped (DB-integration tests skip without Postgres); CI runs lint, migrations, and tests on PostgreSQL 17. The Streamlit dashboard renders SQI-coloured waveforms (< 0.3 red, < 0.6 amber), a quality/contact/motion/confidence timeline, feature tables, anomaly-score charts with a 0.8 review line and `top_deviant` hover, spectral-frame inspection, and the quantum expander behind a `SIMULATED` banner with an automatic warning when `data_class  $\neq$  real`.

12 Security, Privacy and Ethics

A STRIDE-lite threat model in the design documents maps each threat class to a control: spoofing $\rightarrow$ hash-only device keys plus per-device revocation; tampering $\rightarrow$ append-only raw plus idempotent batch keys; repudiation $\rightarrow$ `batch_id`

provenance and device events; information disclosure → subject pseudonymisation, `consent_ref`, planned RLS on subject-facing reads, and keys stored only as SHA-256; denial of service → token-bucket rate limiting, the 2 MB payload cap, and bounded list fields; elevation of privilege → schema-enforced channel registry and a 501-until-JWKS human-auth path that fails closed. Two properties matter most: the server never stores a credential it can lose, and it never stores a number it cannot also describe (provenance, quality, clock state).

Ethically, three lines hold: no medical-device claims anywhere in schema, API, or dashboard; simulated and synthetic data are schema-separated from real data rather than separated by convention; and evaluation reports state when labels are injected. The codebase is source-available for review; it is not a deployed clinical product and has collected no human-subject data.

13 Organisation and Roadmap

VitalQuant is a London-based research startup (team: R. Diwan, CEO; M. Man, CTO; A. Law, CFO) building the platform described here; the closed-weight LAT-D-VQt-1 clinical model is post-trained for the platform by LatticeAG. The staged engineering roadmap moves M0 (platform skeleton; complete) → M1 (`hw_v0` bench collection and ingest soak; verify ring-buffer losslessness, clock-jump handling, and rate-limit behaviour under real radios) → M2 (Phase-A evaluation on the public corpora of Table 1) → M3 (`hw_v1` PCB bring-up plus personal-baseline scoring live on real streams) → M4 (clinical-data partnerships enabling honest-label L4 models) → M5 (quantum-readout bench experiments, conditional on the Section 9 boundary). Each stage is gated by the previous stage’s honesty criteria — the roadmap is a dependency graph, not a pitch deck.

14 Limitations and Conclusion

Limitations: (1) all accuracy figures are against synthetic or injected ground truth — clinical correlation is future work gated behind Phase A/B; (2) the SpO_2 proxy is uncalibrated by design and must never be read as a saturation; (3) PRV approximates, but is not, HRV, and the two diverge under ectopy and motion; (4) the quantum readout is a simulation argument and its detection-statistic layer is under refinement; (5) LAT-D-VQt-1 is closed-weight, externally post-trained, and unmeasured; (6) no human-subject data has been collected, so every clinical framing in this paper is motivational, not evidentiary; (7) the synthetic generator, however faithful in envelope semantics, cannot reproduce the full artefact zoo of free-living wear — Phase-B KS alignment exists to measure exactly that gap.

A fair reading of this paper is therefore not ‘a sepsis detector exists’ but ‘a measurement stack exists that can tell when it is lying’. We regard that as the correct prior deliverable: the field’s reproducibility problems have been caused less by weak models than by unmeasured measurement.

VitalQuant’s thesis is that early-detection wearables must

earn trust through measurement infrastructure — provenance, quality gating, honest labels, and self-incriminating simulation — before they earn clinical attention. The platform implements that thesis end to end, and its own instruments report both where the physics helps ($2.7\times$ ideal squeezed advantage) and where it does not ($\approx 1.14\times$ under dominant tissue noise). We release it as a scientific artefact and invite replication of both results.

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